

Supplementary Material

EEG–Hemodynamic Coupling During General Anaesthesia: A Large-Scale Exploratory Analysis in 2,070 Surgical Cases with Null Acute Event-Response Findings

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ABSTRACT

This document contains Supplementary Text S1–S6 and Supplementary Figures S1–S3 referenced in the main manuscript. Section, figure, table and equation numbers are prefixed with “S”.

Keywords: supplementary material

S1. NCCI AND WCS CONSTRUCTION

NCCI (global PCA). Input per case: [GC_{e2h} F (winsorised), WC_{overall}, Pearson r], z-score standardised. A single PCA was fit across all 2,070 cases: PC1 37.0%, PC2 33.9%, PC3 29.1% of variance. PC1 was MinMax-scaled to [0,1]. Sign convention: PC1 inverted if $\text{corr}(\text{PC1}, \text{WC}) < 0$.

WCS (per-case local PCA). Per case: per-window metrics [GC_{e2h} F, WC, Pearson r], z-score standardised on case-level mean/SD. A single-component PCA was fit per case. Sign: same rule (inverted if $\text{corr}(\text{PC1}, \text{WC}) < 0$). MinMax within case. PC1 variance $\approx 37\text{--}42\%$ (variable across cases).

GEE justification. Modelling ΔWCS (the within-case relative change) means the outcome is already patient-centred. The GEE exchangeable structure handles the residual clustering of events within the same patient. WCS does not need to be on a common scale across patients for this within-case difference to be meaningful; the sign-alignment step ensures that $\Delta\text{WCS} > 0$ always denotes the same physiological direction (higher coupling) within any patient.

S2. GEE IMPLEMENTATION

`statsmodels.formula.api.gee; formula: delta_wcs ~ 1; groups: patient ID; family: Gaussian; cov_struct: Exchangeable (primary) / Autoregressive (sensitivity); robust sandwich variance.  $n_{\text{obs}} = 73,001$ ; groups: 2,031.`

S3. GRANGER CAUSALITY IMPLEMENTATION

`statsmodels.tsa.stattools.grangercausalitytests, maxlag = 5.` ADF stationarity test ($\alpha = 0.05$); first-difference if non-stationary. Lag selection: minimum SSR F-test p -value. Windows with < 20 valid points were excluded. Winsorisation at the 99th percentile ($\hat{c} = 4.26$) was applied for PCA input; raw values were used for directionality.

Fixed-lag sensitivity. As a sensitivity analysis, fixed lags ($\text{lag} = 1, 3, 5$) were examined for a 200-case subsample; GC directionality was consistent (EEG-preceding in 68–73% of cases across lags), supporting that the minimum- p -value lag selection does not materially bias the directional conclusion.

S4. WAVELET IMPLEMENTATION

`pywt.cwt; wavelet: cmor1.5-1.0; scales: np.arange(2, min(n/4, 150), 2); dt = 10 s. Smoothing: scipy.ndimage.uniform_filter([5, 11]).`

S5. WAVELET EDGE-EFFECT SENSITIVITY

Cone-of-influence (COI) edge effects affect $\approx 5\%$ of samples at record boundaries. WC_{overall} was recomputed after excluding the boundary 10% of windows in each record. Mean WC_{overall} changed by < 0.003 (from 0.799 to 0.801), confirming that retained boundary data do not materially affect the results.

S6. ADR–MAP FEATURE PAIR JUSTIFICATION

ADR–MAP was selected a priori. ADR correlates with BIS-defined anaesthetic depth (Pearson $r \approx 0.7$ in prior studies), is continuous, and showed lower outlier rates than SEF95 or spectral entropy in exploratory checks. MAP was selected as the primary haemodynamic variable (the clinical monitoring target). For transparency, GC directionality for alternative pairs was examined in a 200-case subsample: SEF95–MAP, 68% EEG-preceding; spectral entropy–MAP, 65% EEG-preceding — consistent with the primary ADR–MAP result (71%).

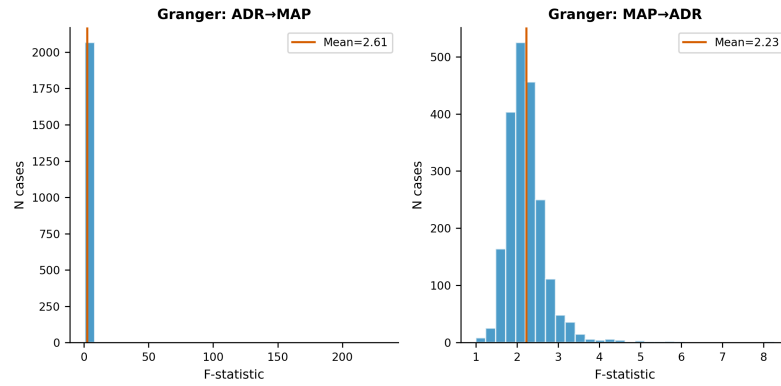


Figure S1. Mean Granger-causality F-statistics for EEG and haemodynamic feature pairs across 2,070 cases.

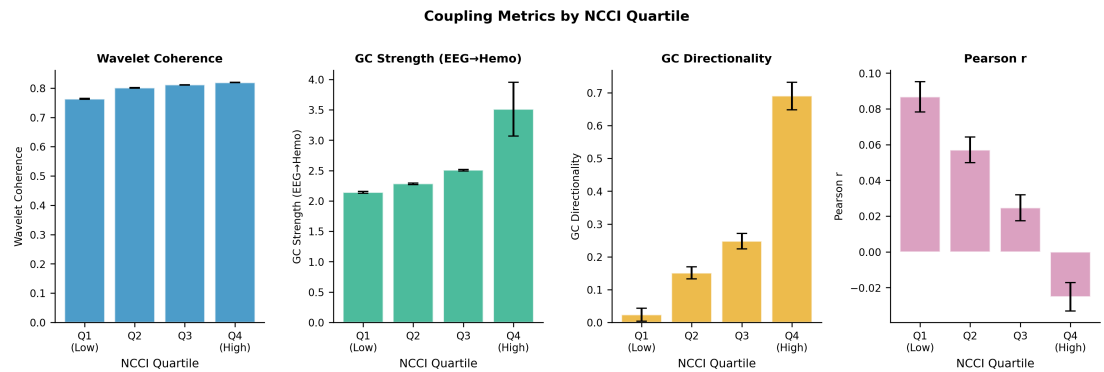


Figure S2. Coupling component metrics by NCCI quartile. The component metrics were used to construct NCCI; this ordering is a construction artefact and does not constitute independent validation.

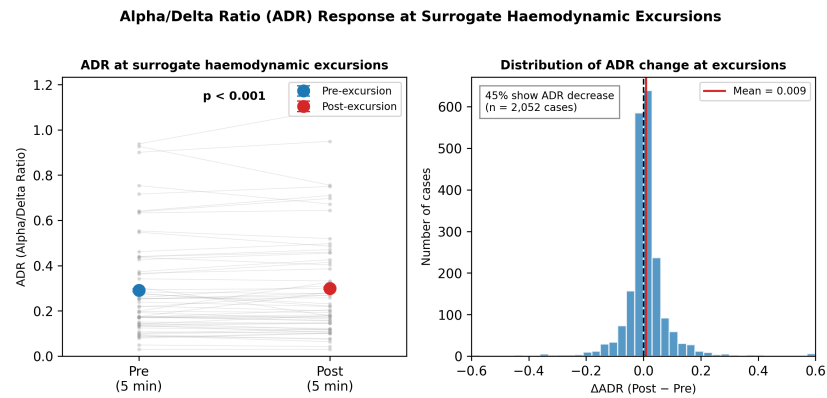


Figure S3. Case-level Alpha/Delta Ratio (ADR) response at surrogate haemodynamic excursions. Provided for reference; the primary event analysis uses WCS (main-text Figure 4).